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1. Project Title

Urban Road Accident Risk and Hotspot Prediction Using Predictive Analytics

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3. Problem Statement

3.1 Problem Overview

Urban road accidents continue to pose a significant challenge to public safety due to increasing traffic density, complex road conditions, and diverse road usage patterns. These accidents result in loss of life, injuries, and economic impact, making road safety a critical concern for urban transportation systems and planning authorities.

With the availability of large-scale road accident datasets, there is an opportunity to move beyond basic statistical analysis and adopt data-driven approaches for understanding accident patterns. Predictive analytics enables the identification of underlying patterns in accident occurrence and severity, providing a foundation for more informed and proactive safety measures.

3.2 Limitations of Conventional Analysis

Traditional road accident analysis primarily focuses on descriptive and retrospective methods, such as analysing accident counts, identifying high-frequency locations, and visualizing past trends. While these approaches are useful for understanding historical data, they suffer from several limitations:

- They do not account for accident severity effectively
- They rely heavily on raw frequency, ignoring contextual factors such as time and environment
- They fail to distinguish between frequent but low-risk areas and less frequent but highly dangerous locations
- They do not support predictive or proactive decision-making

Additionally, real-world accident datasets are inherently imbalanced, with severe and fatal accidents forming only a small proportion of the data. This imbalance often leads to biased models that fail to accurately identify high-risk scenarios.

3.3 Problem Definition

The core problem addressed in this project is the lack of an integrated, predictive framework that combines accident severity, frequency, and contextual factors to identify truly high-risk locations.

This project aims to develop a predictive analytics-based approach that:

- Estimates accident severity using historical data
- Incorporates temporal and environmental features such as time, weather, and lighting conditions
- Addresses class imbalance in accident severity data
- Identifies accident hotspots based on a combination of severity and frequency rather than frequency alone

By shifting from purely descriptive analysis to predictive and risk-based modelling, the project enables a more accurate identification of dangerous locations and supports proactive road safety planning.

3.4 Key Objective of the Problem

The objective of this problem is to move beyond simple accident count analysis and develop a system that can:

- Predict accident severity levels
- Analyse when and under what conditions accidents become more severe
- Identify geographically significant high-risk zones (hotspots)
- Provide meaningful insights that distinguish between frequent and dangerous accident locations

This enables a more realistic and practical understanding of accident risk, supporting better decision-making in urban road safety management.

4. Methodology

4.1 Overview of Approach

The methodology adopted in this project follows a structured predictive analytics pipeline that transforms raw accident data into meaningful insights for accident severity prediction and hotspot identification.

The overall approach consists of the following stages:

- Data acquisition and preprocessing

- Feature engineering
- Model development and comparison
- Handling class imbalance
- Final model training and prediction
- Hotspot identification using risk scoring
- Analytical evaluation through spatial, temporal, and environmental analysis

This structured workflow ensures that both predictive modelling and interpretability are achieved effectively.

Figure 1: Methodology Workflow for Accident Risk Prediction and Hotspot Identification

4.2 Data Preprocessing

The dataset was initially cleaned and filtered to retain only relevant attributes such as accident severity, date, time, location (latitude and longitude), speed limit, road type, lighting conditions, weather conditions, road surface conditions, and day of the week.

Missing values in critical features such as latitude and longitude were removed to ensure accurate spatial analysis. Date and time attributes were converted into appropriate formats, and additional temporal features such as hour, day, and month were extracted.

This preprocessing step ensured that the dataset was consistent, structured, and suitable for both modelling and analysis.

4.3 Feature Engineering

To improve model performance and interpretability, several meaningful features were derived from the raw data:

- Temporal features: hour of the day, day, and month
- Spatial features: latitude and longitude
- Contextual features: weather conditions, lighting conditions, road surface conditions, and speed limit

These features capture different aspects of accident occurrence, including when, where, and under what conditions accidents take place.

4.4 Model Development and Comparison

Multiple supervised machine learning models were implemented and evaluated for accident severity prediction:

- Logistic Regression
- Random Forest (baseline, imbalanced data)
- XGBoost
- XGBoost with oversampling
- Random Forest with oversampling (final model)

The models were trained using a train-test split strategy with stratification to preserve class distribution. Performance was evaluated using classification metrics such as precision, recall, F1-score, and confusion matrix.

This comparative approach helped identify the most suitable model for handling the characteristics of accident data.

4.5 Handling Class Imbalance

Accident severity data is inherently imbalanced, with slight accidents forming the majority class. To address this, an oversampling strategy was applied:

- Fatal and serious accident classes were upsampled
- All classes were balanced to the size of the majority class

This ensured that the model learned to identify severe accident cases more effectively, rather than being biased towards the majority class.

4.6 Final Model Selection and Training

Based on experimental evaluation, the Random Forest model trained on the balanced dataset was selected as the final model.

The model was then retrained on the entire balanced dataset to maximise learning from available data. This trained model was used to predict accident severity (referred to as predicted risk) for all records in the dataset.

4.7 Hotspot Identification Using Risk Score

To identify accident hotspots, a grid-based spatial aggregation approach was used:

- Latitude and longitude values were rounded to create spatial bins
- For each location, the following were computed:
    - Average predicted risk
    - Accident frequency

A composite risk score was defined as:

$$\text{Risk Score} = (4 - \text{Average Predicted Risk}) \times \text{Normalized Accident Count}$$

This formulation ensures that locations with both high severity and high frequency are prioritised.

Locations with very low accident counts were filtered out, and a percentile-based threshold (95th percentile) was used to identify high-risk hotspots.

4.8 Analytical Evaluation

To interpret model outputs and extract meaningful insights, the following analyses were performed:

- Spatial Analysis: Identification and visualization of accident hotspots and geographic clustering
- Temporal Analysis: Examination of accident frequency and severity across different hours, including weekday vs weekend and day vs night comparisons
- Environmental Analysis: Analysis of how weather, road surface, and lighting conditions influence accident severity
- Feature Importance Analysis: Identification of key factors influencing accident severity using model-derived importance scores

These analyses provide a comprehensive understanding of accident patterns beyond predictive modelling.

4.9 Dataset Scope Clarification

The dataset used in this study contains accident records from across Great Britain, including both urban and rural regions. However, the analysis is oriented towards patterns commonly observed in urban environments, where accident density is typically higher.

While explicit filtering of urban-only records was not performed, the insights derived reflect general road safety trends and are applicable to urban contexts.

5. Data Description

5.1 Dataset Source

The dataset used in this project is obtained from the official UK Government open data portal:

Department for Transport (DfT) – Road Safety Open Dataset (STATS19)

This dataset provides detailed records of road collisions in Great Britain, including information about accident severity, location, time, and environmental conditions. The data is collected through the STATS19 collision reporting system and includes all personal injury accidents reported to the police.

The dataset used corresponds to collision records from the most recent five-year period.

The dataset is publicly accessible at:

https://data.dft.gov.uk/road-accidents-safety-data/dft-road-casualty-statistics-collision-last-5-years.csv

5.2 Dataset Overview

The dataset contains:

- 503,475 records (rows)
- 44 attributes (columns)

Each record represents a single road accident along with associated contextual and environmental information.

The dataset covers accident records across Great Britain, including both urban and rural areas.

5.3 Dataset Scope

The analysis in this project is performed on the complete dataset without explicitly filtering only urban records. Using the full dataset allows the model to learn from a wider range of accident conditions and improves its generalization.

The insights derived from the analysis, particularly those related to accident hotspots, temporal patterns, and environmental factors, are applicable to urban road safety where accident density is typically higher.

5.4 Selected Features for Analysis

From the original dataset, a subset of relevant features was selected based on their importance for accident severity prediction.

The features used are:

- collision_severity - Severity level of the accident (target variable)
- date, time - Temporal information
- day_of_week - Day of occurrence
- latitude, longitude - Geographic location

- speed_limit - Speed limit at the accident location
- road_type - Type of road
- light_conditions - Lighting conditions
- weather_conditions - Weather conditions
- road_surface_conditions - Road surface condition
- urban_or_rural_area - Area classification

These features capture temporal, spatial, and environmental aspects influencing accident risk.

5.5 Attribute Encoding and Mappings

The dataset is provided in a coded format where numerical values represent categorical meanings. Proper interpretation of these values is essential.

Table 1: Accident Severity Classification (Target Variable)

Value	Label
1	Fatal
2	Serious
3	Slight

Important Note: Lower values indicate higher severity, while higher values indicate less severe accidents. This relationship is important for interpreting model predictions and analysis results.

Table 2: Weather Condition Classification

Value	Label
1	Fine (no high winds)
2	Raining (no high winds)
3	Snowing (no high winds)
7	Fog or mist
8	Other
9	Unknown
-1	Missing data

Table 3: Road Surface Condition Classification

Value	Label
1	Dry
2	Wet or damp
3	Snow
4	Frost or ice
5	Flood
9	Unknown
-1	Missing

Table 4: Light Condition Classification

Value	Label
1	Daylight
4	Darkness (lights lit)
5	Darkness (lights unlit)
6	Darkness (no lighting)
7	Darkness (unknown lighting)
-1	Missing

5.6 Data Preprocessing Summary

The dataset was preprocessed as follows:

- Records with missing latitude and longitude values were removed
- Date and time fields were converted into appropriate formats
- Temporal features such as hour, day, and month were extracted
- Relevant features were selected for modelling

5.7 Class Imbalance

The dataset exhibits class imbalance in accident severity:

- Slight accidents form the majority
- Serious accidents are fewer
- Fatal accidents are rare

This imbalance reflects real-world conditions and was addressed during model development using oversampling techniques.

6. Implementation

The implementation of the proposed system was carried out using Python in a Google Colab environment. A structured predictive analytics pipeline was followed, consisting of data preprocessing, feature engineering, model development, risk estimation, and hotspot identification.

The implementation is based on the UK Department for Transport (DfT) road accident dataset, which contains historical accident records along with temporal, spatial, and environmental attributes. The dataset was processed to extract relevant features and prepare it for predictive modelling.

Multiple supervised machine learning models, including Logistic Regression, Random Forest, and XGBoost, were developed and compared to evaluate their performance in predicting accident severity. Special attention was given to handling class imbalance present in the dataset using oversampling techniques.

The final model was used to estimate accident risk levels, which were further aggregated spatially to identify potential accident hotspots. In addition, temporal and environmental analyses were performed to understand patterns influencing accident risk.

The implementation steps are described in detail in the following subsections.

6.1 Data Loading and Feature Selection

The dataset was loaded using the Pandas library from the UK Department for Transport (DfT) road accident dataset. From the complete dataset, only relevant attributes related to accident severity, temporal information, location, and environmental conditions were selected for further analysis.

To ensure the validity of spatial analysis, records with missing latitude and longitude values were removed.

```
import pandas as pd


df = pd.read_csv("dft-road-casualty-statistics-collision-last-5-years.csv",
low_memory=False)


df = df[['collision_severity', 'date', 'time',

         'day_of_week', 'latitude', 'longitude',
```

```
         'speed_limit', 'road_type',
         'light_conditions', 'weather_conditions',
         'road_surface_conditions', 'urban_or_rural_area']]


df = df.dropna(subset=['latitude', 'longitude'])
```

6.2 Feature Engineering

Temporal features were extracted from the date and time attributes to capture time-based accident patterns. These include hour of the day, day of the month, and month of occurrence, which are useful for analysing temporal variations in accident risk.

```
df['date'] = pd.to_datetime(df['date'], dayfirst=True, errors='coerce')
df['hour'] = pd.to_datetime(df['time'], format='%H:%M',
errors='coerce').dt.hour
df['day'] = df['date'].dt.day
df['month'] = df['date'].dt.month
```

6.3 Class Imbalance Analysis

The distribution of accident severity classes was analysed to understand the imbalance present in the dataset. Accident severity is categorized into three levels: Fatal, Serious, and Slight.

This analysis is important because real-world accident datasets are typically imbalanced, with severe cases occurring less frequently than minor accidents. Such imbalance can affect the performance of predictive models, particularly in identifying high-risk cases.

```
severity_map = {
    1: "Fatal",
    2: "Serious",
    3: "Slight"
}


severity_labels = df['collision_severity'].map(severity_map)


severity_labels.value_counts()
```

```
severity_labels.value_counts(normalize=True)
```

```
import matplotlib.pyplot as plt


severity_labels.value_counts().plot(kind='bar')
plt.title("Accident Severity Distribution")
plt.xlabel("Severity")
plt.ylabel("Number of Accidents")
plt.xticks(rotation=0)
plt.show()
```

6.4 Dataset Preparation and Train-Test Split

The dataset was prepared for modelling by separating the input features and the target variable. The target variable is accident severity, while all other selected attributes were used as input features.

Unnecessary columns such as date and time were removed, as relevant temporal features had already been extracted during feature engineering. The dataset was then split into training and testing sets using a stratified approach to preserve the original class distribution.

```
from sklearn.model_selection import train_test_split


df_base = df.drop(columns=['date', 'time'])


X = df_base.drop(columns=['collision_severity'])
y = df_base['collision_severity']


X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)
```

6.5 Baseline Model Development (Imbalanced Dataset)

To evaluate the effectiveness of different machine learning approaches, multiple classification models were trained on the original imbalanced dataset. Accident severity prediction is formulated as a multi-class classification problem, and this step provides a baseline for comparison before applying any imbalance handling techniques.

Logistic Regression

Logistic Regression was used as a baseline linear model for multi-class classification.

```
from sklearn.linear_model import LogisticRegression

from sklearn.metrics import classification_report


lr = LogisticRegression(max_iter=200, n_jobs=-1)

lr.fit(X_train, y_train)

y_pred_lr = lr.predict(X_test)


classification_report(y_test, y_pred_lr, zero_division=0)
```

Random Forest

Random Forest, an ensemble-based model, was used to capture non-linear relationships in the data.

```
from sklearn.ensemble import RandomForestClassifier


rf_base = RandomForestClassifier(n_estimators=100, random_state=42,
n_jobs=-1)

rf_base.fit(X_train, y_train)

y_pred_rf = rf_base.predict(X_test)


classification_report(y_test, y_pred_rf, zero_division=0)
```

XGBoost

XGBoost was used as a gradient boosting model for improved predictive performance.

```
from xgboost import XGBClassifier


y_train_xgb = y_train - 1
y_test_xgb = y_test - 1


xgb = XGBClassifier(
    n_estimators=200,
    max_depth=6,
    learning_rate=0.1,
    random_state=42,
    n_jobs=-1
)


xgb.fit(X_train, y_train_xgb)
y_pred_xgb = xgb.predict(X_test)
y_pred_xgb = y_pred_xgb + 1


classification_report(y_test, y_pred_xgb, zero_division=0)
```

6.6 Handling Class Imbalance (Oversampling)

To address the class imbalance present in the dataset, oversampling was applied to the minority classes (Fatal and Serious accidents). The majority class (Slight accidents) was used as the reference, and the minority classes were upsampled to match its size.

This approach ensures that all classes are equally represented during model training, improving the model's ability to correctly identify high-risk (severe) accident cases.

```
from sklearn.utils import resample


df_fatal = df_base[df_base['collision_severity'] == 1]
df_serious = df_base[df_base['collision_severity'] == 2]
df_slight = df_base[df_base['collision_severity'] == 3]
```

```
max_size = len(df_slight)


df_fatal_up = resample(df_fatal, replace=True, n_samples=max_size,
random_state=42)

df_serious_up = resample(df_serious, replace=True, n_samples=max_size,
random_state=42)


df_over_balanced = pd.concat([df_fatal_up, df_serious_up, df_slight])
```

The balanced dataset was then prepared by separating features and target variables, followed by a stratified train-test split.

```
X_over_bal = df_over_balanced.drop(columns=['collision_severity'])

y_over_bal = df_over_balanced['collision_severity']


X_train_ob, X_test_ob, y_train_ob, y_test_ob = train_test_split(

    X_over_bal, y_over_bal,

    test_size=0.2,

    random_state=42,

    stratify=y_over_bal

)
```

6.7 Model Training on Balanced Dataset and Final Model Selection

After addressing class imbalance through oversampling, the models were retrained using the balanced dataset. This step helps improve the model's ability to correctly classify minority classes, particularly severe accidents.

XGBoost (Balanced Dataset)

XGBoost was retrained on the balanced dataset to evaluate its performance after handling class imbalance.

```
y_train_xgb = y_train_ob - 1

y_test_xgb = y_test_ob - 1
```

```
xgb_bal = XGBClassifier(
    n_estimators=200,
    max_depth=6,
    learning_rate=0.1,
    random_state=42,
    n_jobs=-1
)

xgb_bal.fit(X_train_ob, y_train_xgb)
y_pred_xgb_bal = xgb_bal.predict(X_test_ob)
y_pred_xgb_bal = y_pred_xgb_bal + 1

classification_report(y_test_ob, y_pred_xgb_bal, zero_division=0)
```

Random Forest (Balanced Dataset – Final Model)

Random Forest was also trained on the balanced dataset. Based on its performance, it was selected as the final model for accident risk prediction due to its stability and ability to handle complex relationships in the data.

```
from sklearn.metrics import confusion_matrix

rf_ob = RandomForestClassifier(
    n_estimators=200,
    max_depth=15,
    min_samples_split=10,
    random_state=42,
    n_jobs=-1
)

rf_ob.fit(X_train_ob, y_train_ob)
y_pred_ob = rf_ob.predict(X_test_ob)

classification_report(y_test_ob, y_pred_ob, zero_division=0)
```

```
confusion_matrix(y_test_ob, y_pred_ob)
```

6.8 Final Model Training and Risk Prediction

After selecting Random Forest as the final model, it was trained on the entire balanced dataset to utilise all available data for learning. This ensures that the model captures patterns from both majority and minority classes effectively.

The trained model was then used to predict accident severity for all records. These predicted severity values were used as a proxy for accident risk, where lower values indicate higher severity and hence higher risk.

```
rf_final_full = RandomForestClassifier(
    n_estimators=200,
    max_depth=15,
    min_samples_split=10,
    random_state=42,
    n_jobs=-1
)

X_full = df_over_balanced.drop(columns=['collision_severity'])
y_full = df_over_balanced['collision_severity']

rf_final_full.fit(X_full, y_full)
```

```
df_base['predicted_risk'] = rf_final_full.predict(
    df_base.drop(columns=['collision_severity'])
)
```

6.9 Spatial Aggregation for Hotspot Analysis

The geographic coordinates were discretized by grouping nearby locations into spatial bins using rounded latitude and longitude values. This aggregation reduces noise in the data and allows analysis at a location level rather than individual accident records.

For each spatial bin, the average predicted risk and total number of accidents were computed to capture both severity and frequency aspects.

```
df_base['lat_bin'] = df_base['latitude'].round(3)
df_base['lon_bin'] = df_base['longitude'].round(3)


hotspot_df = df_base.groupby(['lat_bin', 'lon_bin']).agg(
    avg_risk=('predicted_risk', 'mean'),
    accident_count=('predicted_risk', 'count')
).reset_index()
```

6.10 Hotspot Identification Using Risk Score

A composite risk score was defined to identify accident-prone locations by combining both accident severity and frequency. The accident count was first normalized to ensure comparability across locations.

The final risk score was computed such that locations with higher severity (lower predicted values) and higher accident frequency receive higher scores. Low-frequency locations were filtered out to avoid noise.

Hotspots were identified using a percentile-based threshold, selecting the top 5% of locations with the highest risk scores.

```
hotspot_df['normalized_count'] = (
    hotspot_df['accident_count'] / hotspot_df['accident_count'].max()
)


hotspot_df['risk_score'] = (
    (4 - hotspot_df['avg_risk']) * hotspot_df['normalized_count']
)


hotspot_df = hotspot_df[hotspot_df['accident_count'] >= 5]
```

```
threshold = hotspot_df['risk_score'].quantile(0.95)


hotspots = hotspot_df[hotspot_df['risk_score'] >= threshold]
```

6.11 Spatial Visualization

Spatial visualization techniques were used to analyse the geographic distribution of accidents and identify high-risk regions. A scatter plot was used to display overall accident locations along with identified hotspots, providing a clear visual distinction between general accident distribution and high-risk areas.

```
import matplotlib.pyplot as plt


plt.figure(figsize=(10, 10))


plt.scatter(
    df_base['longitude'],
    df_base['latitude'],
    s=1,
    alpha=0.1
)


plt.scatter(
    hotspots['lon_bin'],
    hotspots['lat_bin'],
    s=40,
    alpha=0.8
)


plt.xlabel("Longitude")
plt.ylabel("Latitude")
plt.title("Accident Distribution with High-Risk Hotspots")


plt.show()
```

Hexbin plots were used to represent accident density and average predicted risk across locations. These visualizations help in understanding both the frequency of accidents and the severity patterns across different regions.

```
plt.figure(figsize=(10, 10))

hb = plt.hexbin(
    df_base['longitude'],
    df_base['latitude'],
    gridsize=100,
    bins='log'
)

plt.colorbar(hb, label='Log(Accident Count)')
plt.xlabel("Longitude")
plt.ylabel("Latitude")
plt.title("Accident Density (Frequency)")

plt.show()
```

```
plt.figure(figsize=(10, 10))

hb = plt.hexbin(
    df_base['longitude'],
    df_base['latitude'],
    C=df_base['predicted_risk'],
    reduce_C_function=np.mean,
    gridsize=100
)

plt.colorbar(hb, label='Average Risk (Lower = More Dangerous)')
plt.xlabel("Longitude")
plt.ylabel("Latitude")
plt.title("Accident Severity Heatmap")
```

```
plt.show()
```

6.12 Temporal Analysis

Temporal analysis was performed to examine how accident occurrence and risk vary across different time periods. Accident frequency was analysed by aggregating the number of accidents by hour of the day, while average predicted risk was computed to understand severity variations over time.

Additional temporal features were derived to compare accident risk between weekends and weekdays, as well as between night-time and daytime conditions.

```
accidents_by_hour = df_base['hour'].value_counts().sort_index()


import matplotlib.pyplot as plt


plt.figure(figsize=(10,5))
accidents_by_hour.plot(kind='bar')
plt.title("Accident Count by Hour")
plt.xlabel("Hour of Day")
plt.ylabel("Number of Accidents")
plt.show()
```

```
risk_by_hour = df_base.groupby('hour')['predicted_risk'].mean()


plt.figure(figsize=(10,5))
risk_by_hour.plot(marker='o')
plt.xticks(range(0,24))
plt.grid(True, linestyle='--', alpha=0.5)


plt.title("Average Risk by Hour")
plt.xlabel("Hour of Day")
plt.ylabel("Average Predicted Risk (Lower = More Severe)")
plt.show()
```

```
df_base['is_weekend'] = df_base['day_of_week'].isin([1, 7]).astype(int)
weekend_risk = df_base.groupby('is_weekend')['predicted_risk'].mean()


df_base['is_night'] = df_base['hour'].apply(
    lambda x: 1 if (x >= 20 or x <= 5) else 0
)
night_risk = df_base.groupby('is_night')['predicted_risk'].mean()
```

6.13 Environmental Analysis

Environmental factors such as weather conditions, road surface conditions, and lighting conditions were analysed to understand their impact on accident risk. The average predicted risk was computed for each category to identify variations in severity across different environmental scenarios.

Categorical values were mapped to meaningful labels to improve interpretability, and the results were visualized using bar plots.

```
weather_risk =
df_base.groupby('weather_conditions')['predicted_risk'].mean()


weather_labels = {
    1: "Fine",
    2: "Rain",
    3: "Snow",
    4: "Fine + Wind",
    5: "Rain + Wind",
    6: "Snow + Wind",
    7: "Fog/Mist",
    8: "Other",
    9: "Unknown",
    -1: "Missing"
}


weather_plot = weather_risk.sort_values()
weather_plot.index = weather_plot.index.map(weather_labels)
```

```
import matplotlib.pyplot as plt


plt.figure(figsize=(8,5))
weather_plot.plot(kind='barh')
plt.title("Average Predicted Risk by Weather (Lower = More Dangerous)")
plt.xlabel("Predicted Risk")
plt.show()
```

```
road_surface_risk =
df_base.groupby('road_surface_conditions')['predicted_risk'].mean()


road_labels = {
    1: "Dry",
    2: "Wet/Damp",
    3: "Snow",
    4: "Ice/Frost",
    5: "Flood",
    6: "Oil/Diesel",
    7: "Mud",
    9: "Unknown",
    -1: "Missing"
}


road_plot = road_surface_risk.sort_values()
road_plot.index = road_plot.index.map(road_labels)


plt.figure(figsize=(8,5))
road_plot.plot(kind='barh')
plt.title("Average Predicted Risk by Road Surface (Lower = More
Dangerous)")
plt.xlabel("Predicted Risk")
plt.show()
```

```
light_risk = df_base.groupby('light_conditions')['predicted_risk'].mean()


light_labels = {
    1: "Daylight",
    4: "Dark - Lit",
    5: "Dark - Unlit",
    6: "Dark - No Lighting",
    7: "Dark - Unknown",
    -1: "Missing"
}


light_plot = light_risk.sort_values()
light_plot.index = light_plot.index.map(light_labels)


plt.figure(figsize=(8,5))
light_plot.plot(kind='barh')
plt.title("Average Predicted Risk by Light Conditions (Lower = More
Dangerous)")
plt.xlabel("Predicted Risk")
plt.show()
```

6.14 Feature Importance Analysis

Feature importance analysis was performed using the final Random Forest model to identify the relative contribution of different features to accident risk prediction. This provides insight into which factors most influence the model's decisions and helps improve interpretability of the predictive framework.

```
feature_importance = pd.Series(
    rf_final_full.feature_importances_,
    index=X_full.columns
).sort_values(ascending=False)
```

```
import matplotlib.pyplot as plt


plt.figure(figsize=(8,5))
feature_importance.head(10).plot(kind='bar')
plt.title("Top Feature Importance")
plt.ylabel("Importance Score")
plt.show()
```

7. Results

This section presents the outcomes obtained from the implementation of the predictive analytics framework for accident risk estimation and hotspot identification. The results include model performance evaluation, hotspot detection outputs, and visualizations derived from spatial, temporal, and environmental analyses.

The outputs are presented in a structured manner to highlight the performance of different models, the effect of class imbalance handling, and the patterns observed in accident risk across various dimensions.

7.1 Dataset Overview

The dataset used for analysis consists of road accident records obtained from the UK Department for Transport. After initial loading, the dataset contained 503,475 records with 44 attributes.

Following feature selection, only relevant attributes related to accident severity, temporal information, location, and environmental conditions were retained. Records with missing spatial information were removed to ensure valid geographic analysis.

After preprocessing, the final dataset used for modelling contained 503,410 records and 12 attributes.

Table 5: Dataset Summary

Description	Value
Initial number of records	503,475
Initial number of attributes	44
Records after cleaning	503,410
Final number of attributes	12

7.2 Class Imbalance Analysis

The distribution of accident severity classes in the dataset is presented in Table 6. The dataset is highly imbalanced, with slight accidents forming the majority of the records, followed by serious and fatal accidents.

Table 6: Class Distribution (Counts and Percentage)

Severity	Count	Percentage (%)
Slight	385,960	76.67
Serious	109,960	21.84
Fatal	7,490	1.49

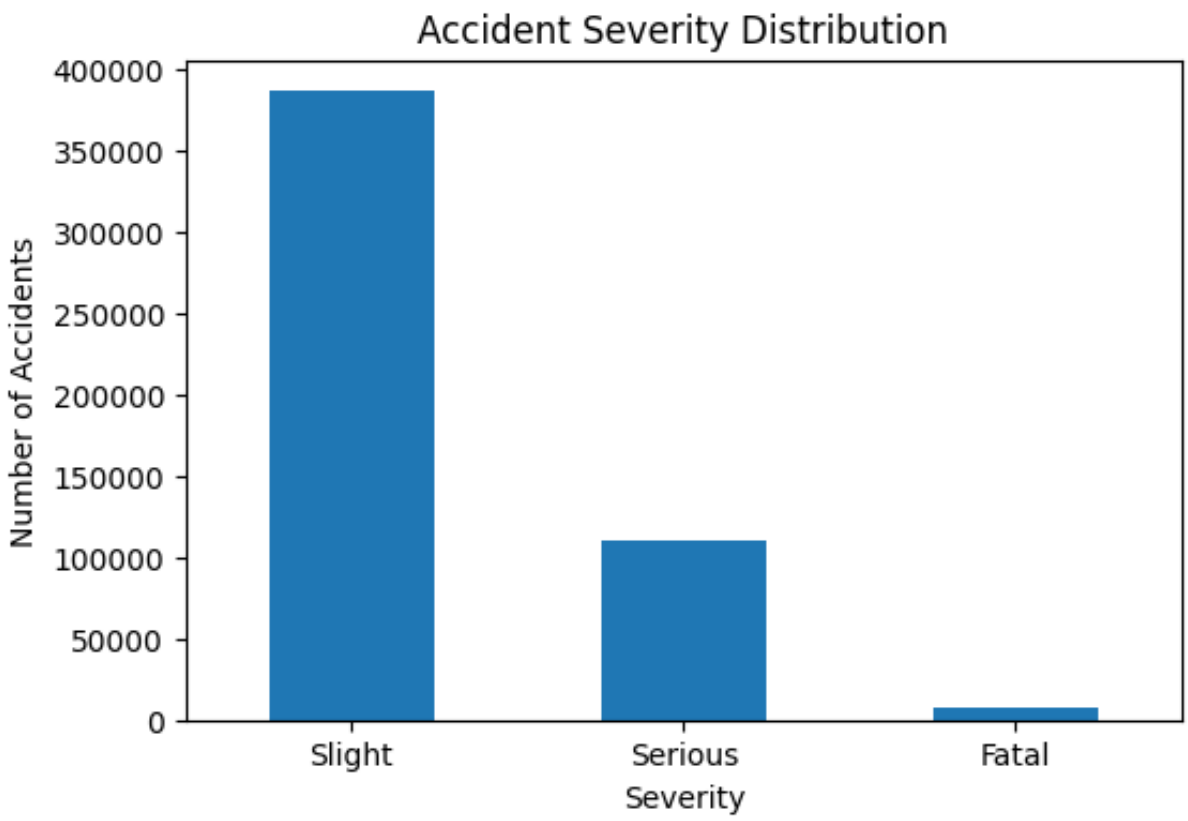

Figure 2: Accident Severity Distribution

7.3 Model Performance (Imbalanced Dataset)

The performance of different machine learning models on the original imbalanced dataset is presented below.

Table 7: Logistic Regression Performance (Imbalanced Dataset)

Class (Severity)	Precision	Recall	F1-Score	Support
Fatal (1)	0.00	0.00	0.00	1,498
Serious (2)	0.00	0.00	0.00	21,992
Slight (3)	0.77	1.00	0.87	77,192
Accuracy			**0.77**	100,682
Macro Avg	0.26	0.33	0.29	100,682
Weighted Avg	0.59	0.77	0.67	100,682

Table 8: Random Forest Performance (Imbalanced Dataset)

Class (Severity)	Precision	Recall	F1-Score	Support
Fatal (1)	0.23	0.00	0.00	1,498
Serious (2)	0.38	0.04	0.08	21,992
Slight (3)	0.77	0.98	0.86	77,192
Accuracy			**0.76**	100,682
Macro Avg	0.46	0.34	0.31	100,682
Weighted Avg	0.68	0.76	0.68	100,682

Table 9: XGBoost Performance (Imbalanced Dataset)

Class (Severity)	Precision	Recall	F1-Score	Support
Fatal (1)	0.00	0.00	0.00	1,498
Serious (2)	0.49	0.02	0.03	21,992
Slight (3)	0.77	1.00	0.87	77,192
Accuracy			**0.77**	100,682
Macro Avg	0.42	0.34	0.30	100,682
Weighted Avg	0.70	0.77	0.67	100,682

7.4 Model Performance (Balanced Dataset)

The performance of models after applying oversampling is presented below.

Table 10: XGBoost Performance (Balanced Dataset)

Class (Severity)	Precision	Recall	F1-Score	Support
Fatal (1)	0.60	0.73	0.66	77,192
Serious (2)	0.48	0.34	0.40	77,192
Slight (3)	0.52	0.57	0.54	77,192
Accuracy			**0.54**	231,576
Macro Avg	0.53	0.54	0.53	231,576
Weighted Avg	0.53	0.54	0.53	231,576

Table 11: Random Forest Performance (Balanced Dataset – Final Model)

Class (Severity)	Precision	Recall	F1-Score	Support
Fatal (1)	0.72	0.85	0.78	77,192
Serious (2)	0.62	0.44	0.52	77,192
Slight (3)	0.58	0.64	0.61	77,192
Accuracy			**0.64**	231,576
Macro Avg	0.64	0.64	0.63	231,576
Weighted Avg	0.64	0.64	0.63	231,576

Table 12: Confusion Matrix for Random Forest (Balanced Dataset)

Actual \ Predicted	Fatal (1)	Serious (2)	Slight (3)
Fatal (1)	65,802	3,890	7,500
Serious (2)	15,235	34,171	27,786
Slight (3)	10,992	17,128	49,072

7.5 Hotspot Identification Results

The results of hotspot identification based on the computed risk score are presented below.

Hotspot Coverage

- Total number of locations analysed: 13,420

- Number of identified hotspots: 698

Table 13: Top 10 High-Risk Hotspots

Latitude	Longitude	Avg Risk	Accident Count	Normalized Count	Risk Score
52.564	-1.503	1.000	22	0.393	1.179
51.461	-0.116	2.870	54	0.964	1.089
51.428	-0.168	2.982	56	1.000	1.018
51.584	-0.072	2.982	55	0.982	1.000
51.571	-0.096	2.981	52	0.929	0.946
53.815	-1.562	2.038	26	0.464	0.911
51.409	-0.122	2.958	48	0.857	0.893
51.750	-1.244	2.886	44	0.786	0.875
53.746	-2.488	2.080	25	0.446	0.857
51.613	-0.065	3.000	47	0.839	0.839

Table 14: Top 10 High-Frequency Areas

Latitude	Longitude	Avg Risk	Accident Count	Normalized Count	Risk Score
51.428	-0.168	2.982	56	1.000	1.018
51.584	-0.072	2.982	55	0.982	1.000
51.461	-0.116	2.870	54	0.964	1.089
51.571	-0.096	2.981	52	0.929	0.946
51.409	-0.122	2.958	48	0.857	0.893
51.613	-0.065	3.000	47	0.839	0.839
51.597	-0.110	3.000	45	0.804	0.804
51.750	-1.244	2.886	44	0.786	0.875
51.525	-0.035	2.977	44	0.786	0.804
51.559	-0.254	3.000	43	0.768	0.768

Correlation Analysis

Correlation between average risk and accident frequency: 0.159

7.6 Spatial Analysis Results

The spatial distribution of accidents and identified hotspots is presented through the following visualizations.

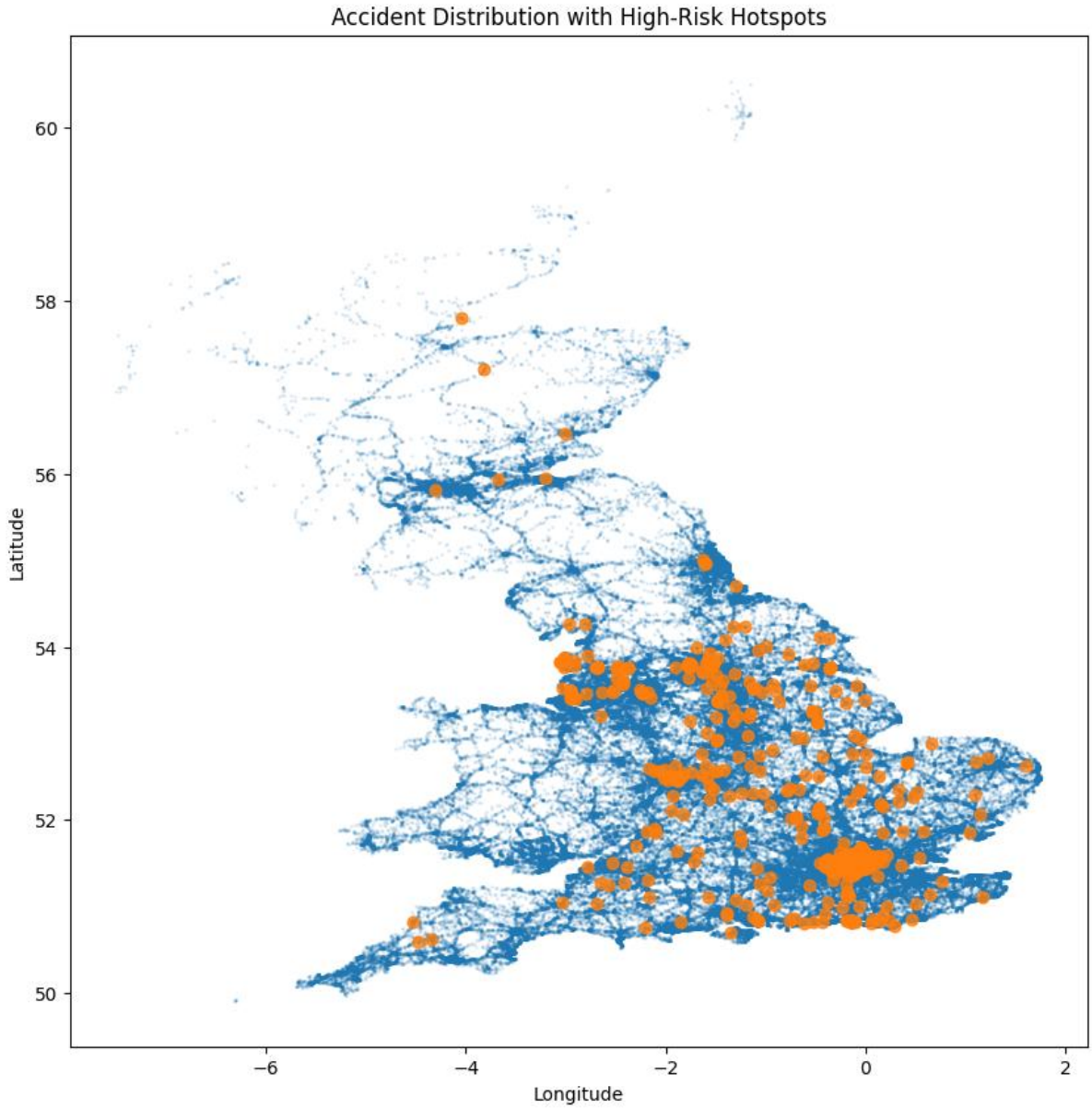

Figure 3: Accident Distribution with High-Risk Hotspots

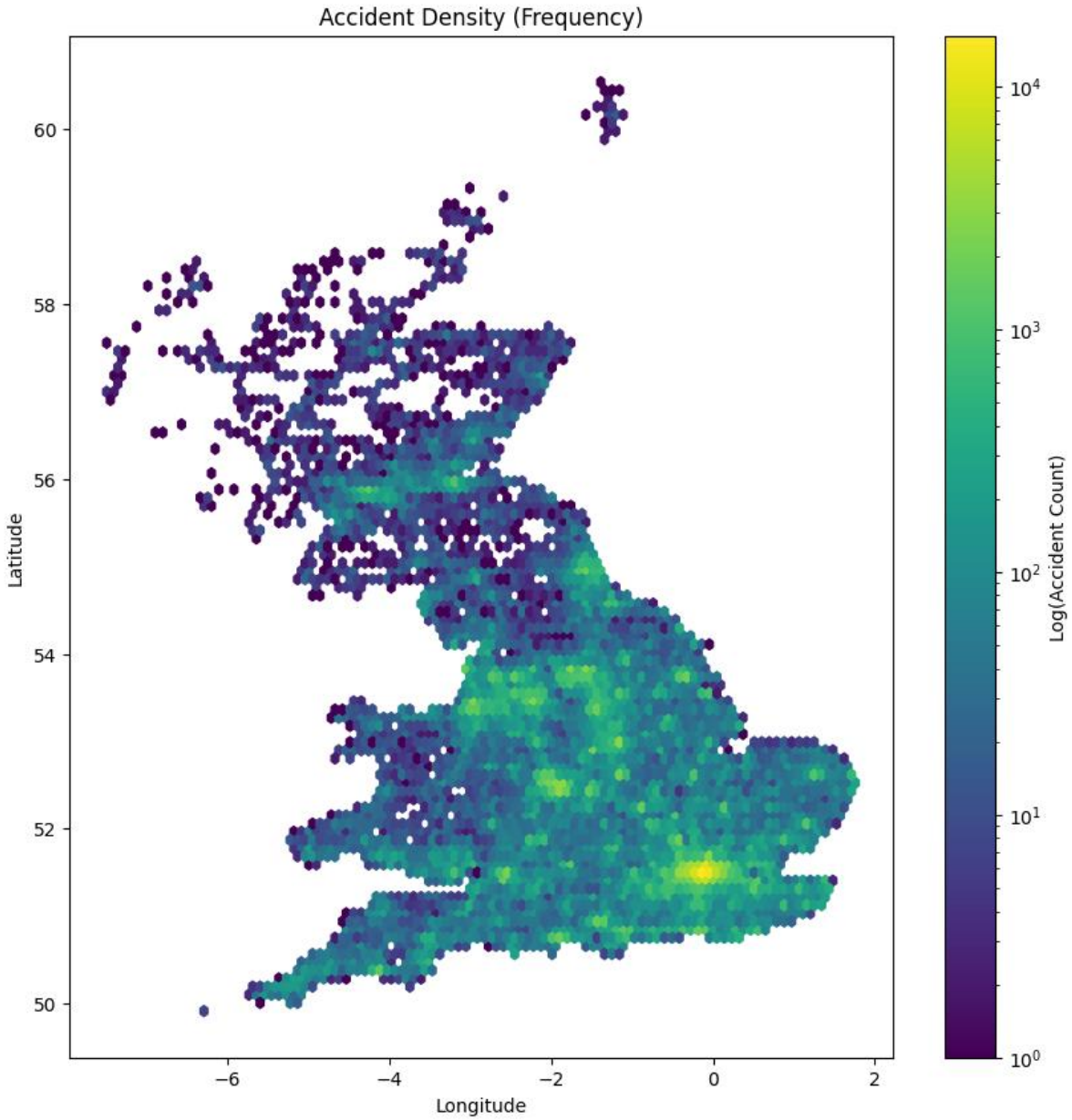

Figure 4: Accident Density (Frequency)

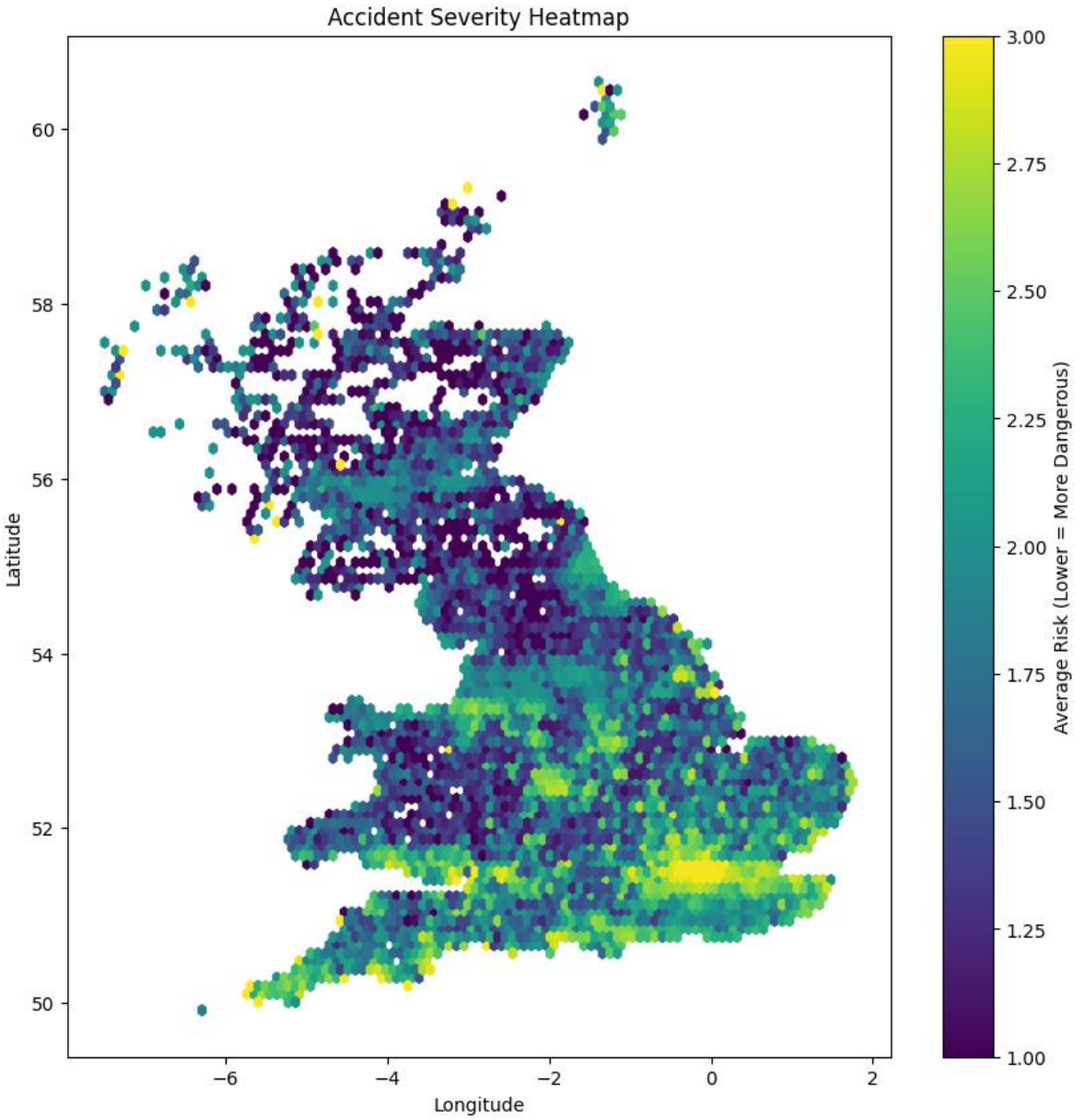

Figure 5: Accident Severity Heatmap

7.7 Temporal Analysis Results

The variation of accident frequency and predicted risk across different time periods is presented below.

Table 15: Accident Count by Hour

Hour	Accident Count
0	7,772

1	5,493
2	3,975
3	3,339
4	3,012
5	4,700
6	9,537
7	20,098
8	31,404
9	21,945
10	22,265
11	25,765
12	29,250
13	30,409
14	32,728
15	41,185
16	41,597
17	43,722
18	35,556
19	27,533
20	20,180
21	16,638
22	14,389
23	10,918

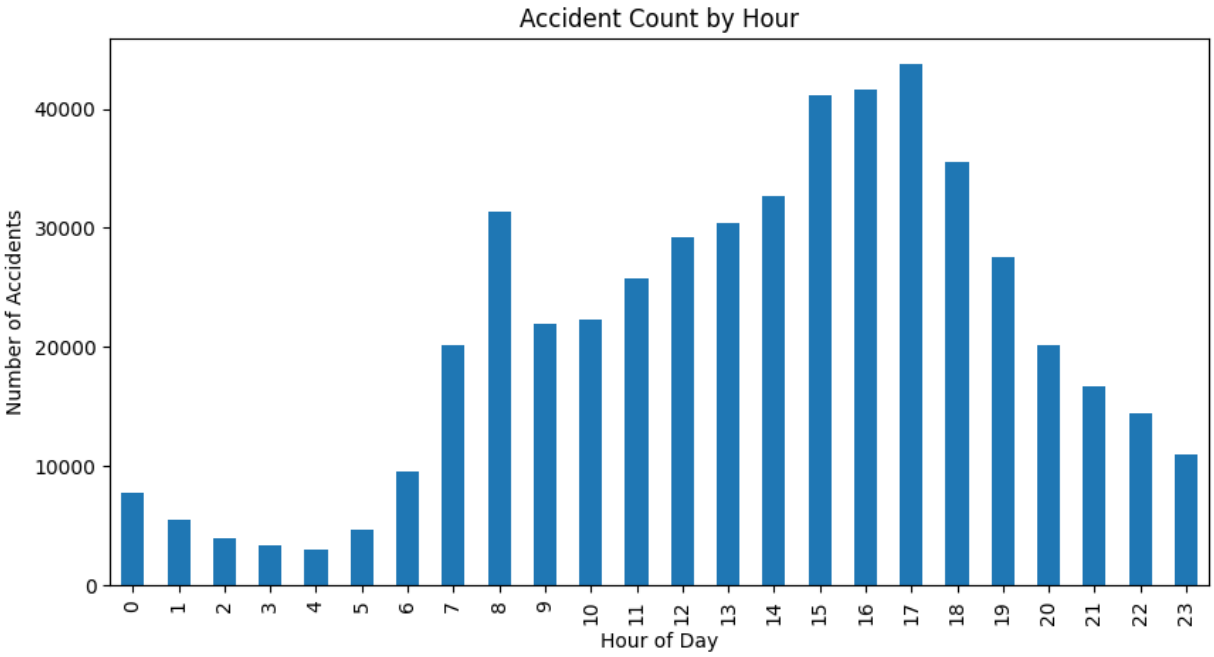

Figure 6: Accident Count by Hour

Table 16: Average Predicted Risk by Hour

Hour	Avg Predicted Risk
0	2.233
1	2.175
2	2.122
3	2.166
4	2.219
5	2.304
6	2.420
7	2.512
8	2.600
9	2.501
10	2.451
11	2.430
12	2.442
13	2.460

14	2.454
15	2.485
16	2.451
17	2.454
18	2.476
19	2.442
20	2.378
21	2.283
22	2.222
23	2.223

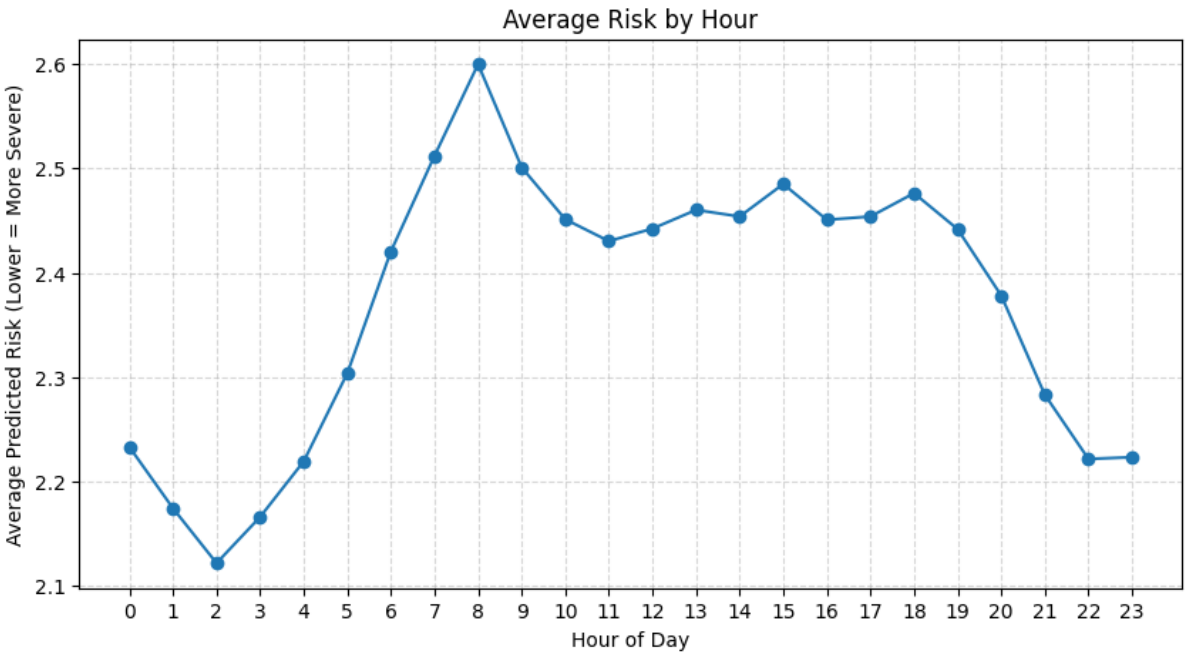

Figure 7: Average Predicted Risk by Hour

Table 17: Weekend vs Weekday Risk

Category	**Avg Predicted Risk**
Weekday (0)	2.457
Weekend (1)	2.367

Table 18: Night vs Day Risk

Category	Avg Predicted Risk
Day (0)	2.471
Night (1)	2.264

7.8 Environmental Analysis Results

The variation of predicted accident risk under different environmental conditions is presented below.

Table 19: Average Predicted Risk by Weather Conditions

Weather Condition	Avg Predicted Risk
Fog/Mist	2.311
Fine + Wind	2.359
Rain + Wind	2.362
Missing	2.385
Fine	2.405
Rain	2.431
Snow	2.609
Snow + Wind	2.668
Other	2.786
Unknown	2.934

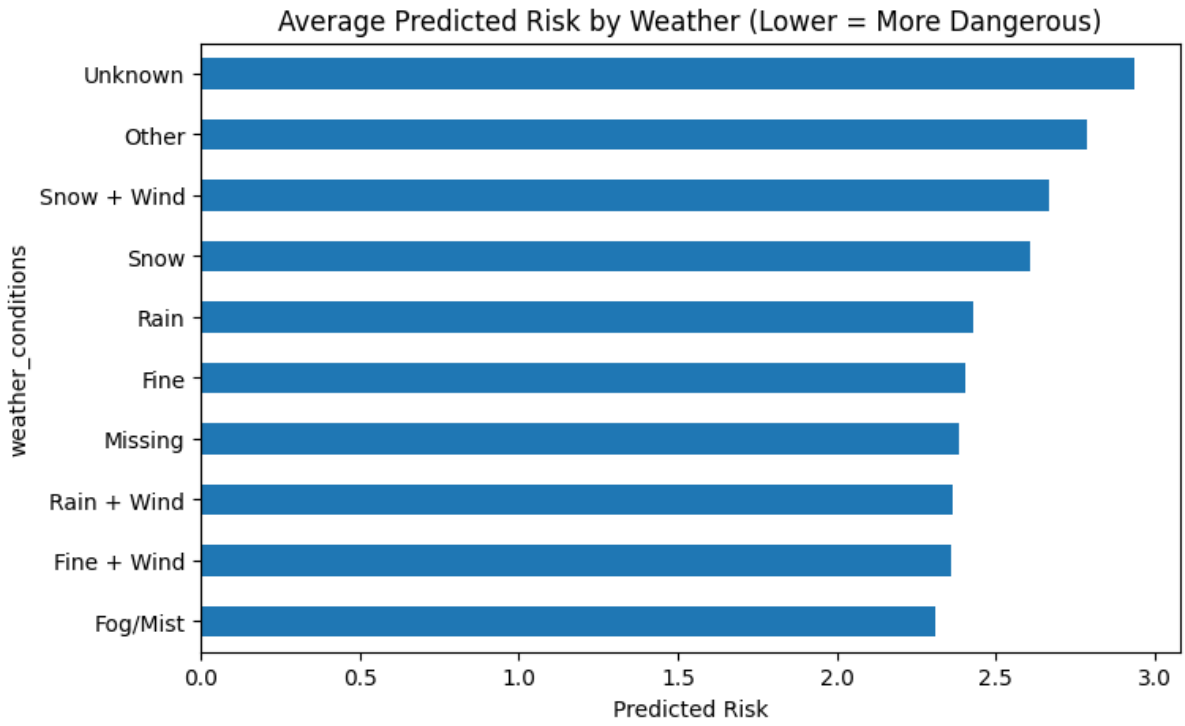

Figure 8: Average Predicted Risk by Weather Conditions

Table 20: Average Predicted Risk by Road Surface Conditions

Road Surface Condition	Avg Predicted Risk
Wet/Damp	2.358
Dry	2.441
Flood	2.663
Snow	2.682
Ice/Frost	2.711
Missing	2.845
Unknown	2.977

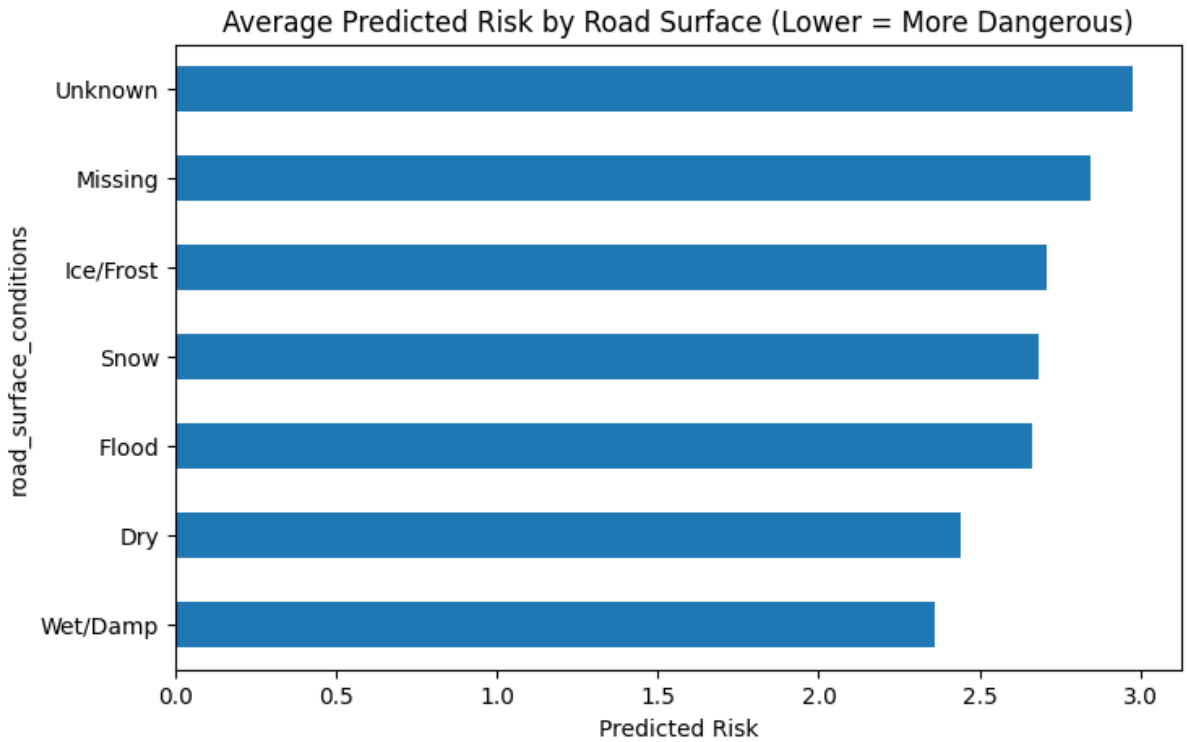

Figure 9: Average Predicted Risk by Road Surface Conditions

Table 21: Average Predicted Risk by Light Conditions

Light Condition	Avg Predicted Risk
Dark – No Lighting	1.670
Missing	2.105
Dark – Unlit	2.380
Dark – Lit	2.454
Daylight	2.478
Dark – Unknown	2.736

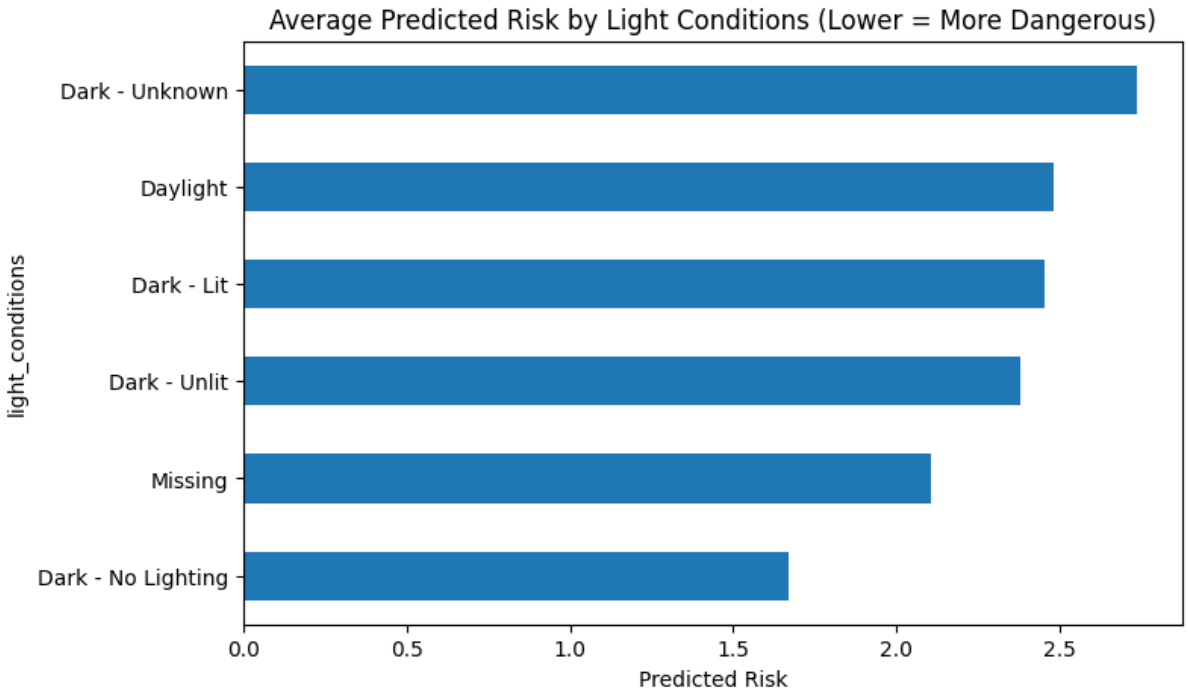

Figure 10: Average Predicted Risk by Light Conditions

7.9 Feature Importance Results

The most influential features contributing to accident risk prediction, as determined by the Random Forest model, are presented below.

Table 22: Top 10 Feature Importance Scores

Feature	Importance Score
Latitude	0.167
Longitude	0.150
Speed Limit	0.146
Hour	0.101
Urban or Rural Area	0.085
Day	0.082
Month	0.060
Light Conditions	0.057
Road Type	0.048

Day of Week	0.047

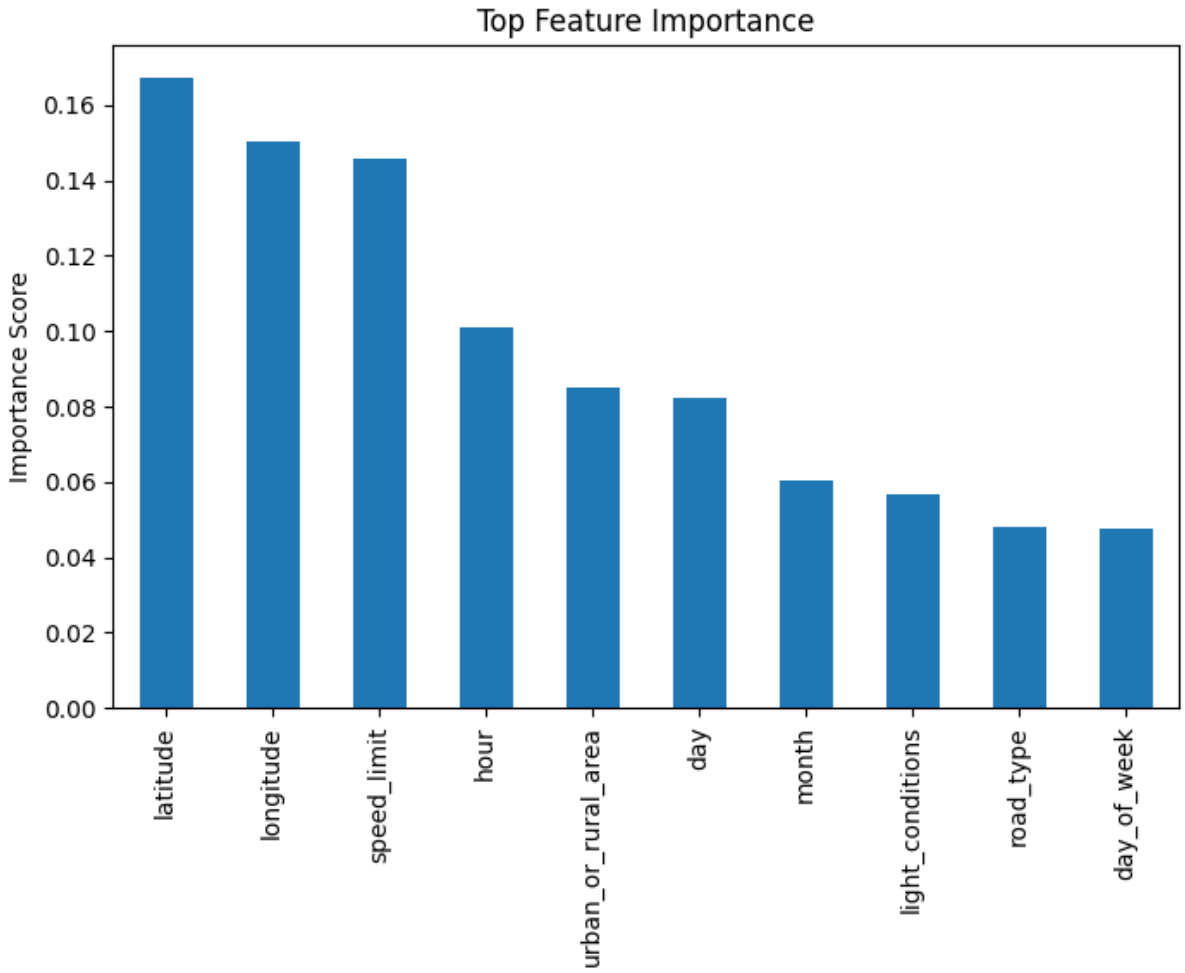

Figure 11: Top Feature Importance

8. Interpretation and Discussion

The results obtained from the predictive analytics framework provide insights into accident severity patterns, model performance, and the influence of spatial, temporal, and environmental factors on accident risk.

This section interprets the observed results by analysing patterns in the dataset, evaluating the effectiveness of different modelling approaches, and examining how various factors contribute to accident risk. The discussion focuses on understanding the behaviour of the models, the impact of class imbalance, and the significance of identified hotspots and influencing features.

The insights derived from this analysis highlight key trends in accident occurrence and severity, and demonstrate the usefulness of predictive analytics in supporting data-driven road safety assessment.

8.1 Dataset Characteristics and Class Imbalance

The dataset used in this study is large-scale and comprehensive, containing over 500,000 accident records. Even after preprocessing and removal of records with missing spatial information, the dataset retains a substantial number of observations, ensuring that the analysis is based on a rich and representative sample of real-world accident scenarios. The reduction in the number of attributes from 44 to 12 reflects a focused selection of features relevant to accident severity prediction, which helps in improving model efficiency without compromising essential information.

The class distribution of accident severity reveals a significant imbalance in the dataset. Slight accidents constitute the majority of the records, while serious and fatal accidents represent a much smaller proportion. This imbalance is expected in real-world traffic data, where minor accidents occur far more frequently than severe or fatal incidents.

Such class imbalance has important implications for predictive modelling. Machine learning models trained on imbalanced data tend to be biased towards the majority class, leading to poor performance in identifying minority classes, particularly severe accidents. This makes it challenging to accurately predict high-risk cases, which are of greater importance in road safety analysis.

Therefore, addressing class imbalance becomes a critical step in the modelling process to ensure that the predictive system can effectively capture and identify severe accident scenarios.

8.2 Model Performance Analysis

The performance of machine learning models on the imbalanced dataset highlights the strong influence of class imbalance on prediction behaviour. Logistic Regression, Random Forest, and XGBoost all achieve relatively high overall accuracy; however, this is largely due to their ability to correctly classify the majority class (slight accidents). All three models show extremely poor performance for fatal and serious accident classes, with near-zero recall in several cases. This indicates that the models are heavily biased towards predicting the majority class and fail to identify high-risk cases effectively.

Logistic Regression performs the weakest among the models, as it is unable to capture complex patterns in the data and completely fails to distinguish minority classes. Random Forest and XGBoost perform slightly better, but still show very low recall for fatal and serious accidents, confirming that imbalance significantly limits their effectiveness.

After applying oversampling, there is a noticeable improvement in the performance of both models. XGBoost demonstrates improved recall for minority classes, indicating that it is better able to recognize severe accidents when class distribution is balanced. However, its overall accuracy decreases, which is expected as the model no longer favors the majority class.

Random Forest shows the most balanced performance after oversampling. It achieves higher recall for fatal accidents and improved performance for serious cases while maintaining relatively stable overall accuracy. This indicates that Random Forest is better suited for handling the trade-off between class balance and predictive performance.

The confusion matrix further supports this observation, showing that the model is able to correctly classify a significant number of fatal and serious cases, while still maintaining reasonable performance for slight accidents. Although some misclassification remains, the overall distribution of predictions is more balanced compared to the imbalanced case.

Overall, the results demonstrate that handling class imbalance is essential for improving the model's ability to detect high-risk accident scenarios, and that Random Forest provides the most reliable performance for this task.

8.3 Hotspot Analysis

The hotspot identification results demonstrate the effectiveness of combining accident severity and frequency into a unified risk score. By incorporating both average predicted risk and normalized accident count, the model is able to identify locations that are not only accident-prone but also potentially more dangerous.

The total number of identified hotspots represents a small proportion of all analysed locations, indicating that high-risk areas are concentrated rather than uniformly distributed. This suggests that accident risk is localized and influenced by specific geographic or infrastructural conditions.

A key observation from the results is the distinction between high-risk hotspots and high-frequency areas. While some locations appear in both categories, others differ significantly. High-frequency areas are characterized by a large number of accidents but may not necessarily have high severity, whereas high-risk hotspots are identified based on a combination of severity and frequency. This highlights the importance of considering both factors when assessing road safety.

The relatively low correlation between average risk and accident frequency further supports this distinction. It indicates that areas with a high number of accidents do not always correspond to areas with more severe accidents. Therefore, relying solely on accident frequency may overlook locations with fewer but more severe incidents.

Overall, the hotspot analysis demonstrates that the proposed risk-based approach provides a more comprehensive identification of critical locations compared to traditional frequency-based methods, enabling a more targeted understanding of accident-prone regions.

8.4 Spatial Patterns

The spatial visualizations reveal that road accidents are not uniformly distributed across the study area, but instead exhibit clear clustering patterns. The scatter plot shows a dense concentration of accident points in specific regions, indicating that certain geographic areas experience a significantly higher number of incidents compared to others.

The density hexbin plot further emphasizes this pattern by highlighting regions with high accident frequency. These dense clusters suggest the presence of urban centres or heavily trafficked road networks where accident occurrence is more frequent. However, the distribution is not continuous, and large areas show relatively low accident density, indicating variability in traffic conditions and infrastructure.

The accident severity heatmap provides additional insight by illustrating how predicted risk varies spatially. Unlike the density plot, the severity distribution is not strictly aligned with accident frequency. Some regions with high accident density do not necessarily exhibit the highest risk levels, while certain lower-density areas show relatively higher severity. This reinforces the observation that accident frequency alone is not sufficient to identify high-risk locations.

The overlay of identified hotspots on the scatter plot highlights that high-risk locations are concentrated within specific regions but are not limited only to the most densely populated areas. This indicates that the proposed risk-based approach is able to capture both severity and frequency, enabling a more nuanced identification of accident-prone regions.

Overall, the spatial analysis demonstrates that accident occurrence and severity vary significantly across locations, and that combining these dimensions provides a more comprehensive understanding of geographic risk patterns.

8.5 Temporal Patterns

The temporal analysis reveals clear variations in accident frequency and severity across different times of the day. The accident count by hour shows that incidents are relatively low during early morning hours, increase significantly during daytime, and peak during late afternoon and early evening hours. This pattern is consistent with typical traffic flow, where higher vehicle density during working hours leads to increased accident occurrence.

In contrast, the average predicted risk follows a different trend. Lower predicted risk values, which indicate more severe accidents, are observed during late night and early morning hours. As the day progresses, the predicted risk values increase, indicating that accidents occurring during daytime tend to be less severe on average. This suggests that although fewer accidents occur at night, they are relatively more severe compared to daytime incidents.

The comparison between weekday and weekend risk shows a slight variation, with weekends exhibiting lower predicted risk values. This indicates that accidents during weekends tend to be relatively more severe than those on weekdays, even though overall traffic volume may differ.

Similarly, the comparison between night and day further supports this observation. Night-time conditions show lower predicted risk values compared to daytime, indicating higher severity during night hours. Factors such as reduced visibility, driver fatigue, and lower traffic regulation during night-time may contribute to this pattern.

Overall, the temporal analysis highlights a key distinction between accident frequency and severity. While accidents are more frequent during peak traffic hours, their severity tends to be higher during off-peak periods such as night-time, emphasizing the importance of considering both dimensions in accident risk analysis.

8.6 Environmental Factors

The environmental analysis highlights the influence of external conditions such as weather, road surface, and lighting on accident severity.

The results for weather conditions show that lower predicted risk values (indicating higher severity) are associated with conditions such as fog or mist and adverse wind conditions. In contrast, conditions like snow and unknown categories exhibit higher predicted risk values, indicating relatively less severe accidents on average. This suggests that reduced visibility conditions may contribute more significantly to accident severity than general weather conditions alone.

The analysis of road surface conditions reveals that wet or damp roads are associated with lower predicted risk values compared to dry roads, indicating relatively higher severity under such conditions. More extreme conditions such as snow, ice, and flood show higher predicted risk values, which may reflect reduced traffic volume or more cautious driving behaviour in such environments. This indicates that commonly encountered conditions like wet roads may pose a higher practical risk compared to less frequent extreme conditions.

Lighting conditions show the most pronounced variation in predicted risk. Accidents occurring in areas with no lighting have the lowest predicted risk values, indicating the highest severity. This is followed by dark and unlit conditions, while daylight conditions show relatively higher predicted risk values, indicating lower severity. This pattern suggests that visibility plays a critical role in accident severity, with poorly lit environments significantly increasing the likelihood of severe outcomes.

Overall, the environmental analysis demonstrates that factors affecting visibility and driving conditions, particularly lighting and road surface, have a strong influence on accident severity. These findings emphasize the importance of considering environmental context when analysing accident risk.

8.7 Feature Importance Interpretation

The feature importance analysis provides insight into the factors that most influence accident risk prediction in the final Random Forest model. Among all features, geographic variables such as latitude and longitude have the highest importance scores, indicating that location plays a dominant role in determining accident risk. This aligns with the earlier spatial analysis, where accident occurrences and hotspots were found to be concentrated in specific regions.

Speed limit emerges as one of the most significant non-spatial features, suggesting that road characteristics and traffic regulations have a notable impact on accident severity. Temporal features such as hour, day, and month also contribute meaningfully, reflecting the influence of time-based patterns observed in the temporal analysis.

Environmental factors such as light conditions and road type show moderate importance, indicating that while they affect accident severity, their impact is comparatively lower than spatial and temporal variables. The presence of both spatial and non-spatial features among the top contributors highlights the multi-dimensional nature of accident risk.

Feature importance analysis is performed at the final stage of the pipeline to interpret the behaviour of the trained model after all preprocessing, balancing, and modelling steps are completed. This ensures that the importance scores reflect the actual decision-making process of the final model used for prediction, rather than intermediate or incomplete models.

It is important to note that feature importance indicates the relative influence of variables in the predictive model, but does not necessarily imply direct causal relationships. Therefore, while features such as speed limit are identified as important, detailed domain-specific analysis of their effects is beyond the scope of this study.

9. Conclusion

This project presented a predictive analytics approach for analysing road accident risk and identifying high-risk locations using machine learning techniques. By leveraging the UK Department for Transport road accident dataset, the study developed a comprehensive pipeline involving data preprocessing, feature engineering, model training, and hotspot detection.

The results demonstrated that class imbalance significantly affects model performance, particularly in predicting severe accidents. The application of oversampling techniques improved the model's ability to identify minority classes, with Random Forest emerging as the most effective model for balanced and reliable prediction.

The integration of predicted accident severity with spatial aggregation enabled the identification of accident hotspots using a composite risk score. This approach provided a more comprehensive understanding of risk by combining both severity and frequency, rather than relying solely on accident counts.

Further analysis of spatial, temporal, and environmental factors revealed meaningful patterns in accident occurrence and severity. The findings highlighted that accident risk varies across locations, time periods, and external conditions, emphasizing the importance of a multi-dimensional analysis framework.

Overall, the study demonstrates the effectiveness of predictive analytics in supporting data-driven road safety assessment and provides a scalable approach for identifying high-risk areas, which can assist in planning targeted interventions and improving traffic safety.

10. Project Repository

https://github.com/yuvaraj-15/urban-road-accident-risk-and-hotspot-prediction-using-predictive-analytics